

Pau Verdejo Gallardo

Applied Statistics and Physics student
Data science and machine learning

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🌐 pauverdejog-ui.github.io/pauverdejog

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🐙 [pauverdejog-ui](https://github.com/pauverdejog-ui)

Education

- 2023–2027 **BSc in Applied Statistics**, *Universitat Autònoma de Barcelona (UAB)*
- 2026 **Erasmus+ exchange**, *Linköping University, Sweden*
Machine Learning for Computer Vision, Detection and Estimation of Signals, Complex Networks and Big Data, Statistical Theory (Advanced Course).
- 2025–now **BSc in Physics**, *UNED (distance learning)*
- 2020–2022 **International Baccalaureate Diploma**, *Àgora Sant Cugat International School*
Bojos per la Física programme for students with an interest in physics (Fundació Catalunya La Pedrera).

Projects

Code, reports and more projects at pauverdejog-ui.github.io/pauverdejog

- 2026 **Higgs boson detection with graph neural networks**, *Machine Learning II, UAB*, individual
GNN (EdgeConv, PyTorch Geometric) that represents each collision as a graph of particles: AUC of 0.857 versus 0.788 for an MLP on one million events, above the published XGBoost benchmark.
- 2026 **Predicting League of Legends wins with Bayesian networks**, *Complex Data Modelling, UAB*, pair project
Naive Bayes and Augmented Naive Bayes (bnlearn, gRain) on minute-10 data from 9,131 games; Markov blanket feature selection, cross-validation and paired tests with Holm correction. Best calibration: Brier score 0.189.
- 2026 **Image classification with convolutional networks**, *Machine Learning for Computer Vision, Linköping University*, team of 5
Controlled study in PyTorch on CIFAR-10 of initialisation, normalisation, augmentation, mixup and linear probing; an ensemble of three networks reaches 84.9% accuracy.
- 2025 **Exoplanet detection with Kepler telescope data**, *Machine Learning, UAB*, team of 3
k-NN, SVM and Random Forest (scikit-learn) to classify 7,015 Kepler signals, with 10-fold cross-validation and paired tests. Final model: AUC of 0.999 and 98.7% test accuracy.

Technical skills

- Methods Supervised and unsupervised machine learning, deep learning (MLP, CNN, GNN), statistical inference, linear and mixed models, Bayesian networks, numerical simulation (Runge–Kutta, Monte Carlo)
- Languages Python, R, C++, MATLAB, SQL, $\text{\LaTeX}$
- Libraries PyTorch, PyTorch Geometric, scikit-learn, XGBoost, pandas, NumPy, Matplotlib, Seaborn; ggplot2, lme4, bnlearn (R)
- Tools Git and GitHub, Docker, Linux, Jupyter, Google Colab, Tableau, Power BI

Experience

- 2026–now **Organiser**, *Barcelona Mathematics League (Liga Matemàtica de Barcelona)*
Organising and running the competition.
- 2022–2025 **Private tutor in mathematics and physics**, *Barcelona*
Lessons for upper secondary (bachillerato) students.
- 2019–2022 **Volunteer**, *International Baccalaureate CAS programme, Barcelona*
Activities for people with disabilities at the Centre d'Alt Rendiment (CAR) high-performance sports centre, and food collection for El Gran Recapte.

Other

- Languages Spanish and Catalan (native), English (C1 Advanced, Cambridge)
- Strengths Curiosity, problem solving, adaptability and teamwork
- Interests Artificial intelligence, photography, video editing, film, theatre, philosophy and history